

ISAAC HASHIMI

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EDUCATION

University of California, Santa Cruz

B.S. Electrical Engineering · Minor: Technology and Information Management

September 2025 – Present
Specialization: Electrical Engineering

Mountain House High School

High School Diploma · Graduated with High Honors – 4.2 GPA

2020 – 2025
Specialization: Computer Science Pathway

SKILLS

Languages: C, Python, Java, Verilog, SystemVerilog

Hardware & CAD: RISC-V architecture, FPGA simulation, RTL design, hardware verification, Onshape (Advanced Part & Surfacing), enclosure design, 2D manufacturing drawings

Tools & Platforms: Git, NVIDIA Nemotron, cloud compute (Brev), Microsoft Office Suite, Google Workspace

Concepts: Pipelined CPU design, multi-agent systems, adversarial test generation, high-current cabling, power electronics

CERTIFICATIONS

Oracle – Java SE 8 Programmer I · AMD – Triton Kernel Development on AMD Instinct GPUs · Onshape – Advanced Part and Surfacing Design

WORK EXPERIENCE

Formula Slug, UC Santa Cruz, Electrical Team Member

October 2025 – Present

- Designed the FSAE charger enclosure in Onshape, modeling geometry for mechanical protection, moisture sealing, electrical isolation, and thermal management; converted to dimensioned 2D manufacturing drawings with tolerances and assembly notes.
- Assembled and routed high-current cabling to FSAE safety standards, ensuring proper insulation, grounding, and thermal considerations throughout the power electronics system.
- Collaborated across electrical, mechanical, and software subteams to troubleshoot system-level issues and maintain wiring diagrams, integration checklists, and assembly documentation.

SkillsUSA, Competition Manager

August 2023 – May 2025

- Automated data entry pipelines to manage registration and logistics across multiple competitive events, ensuring accurate records and on-time execution.
- Coordinated with advisors, judges, and competitors as a primary point of contact, demonstrating reliability and professionalism under deadline pressure.

Mountain House High School, Summer Tutor

June 2023 – August 2023

- Tutored students in mathematics, computer science, and programming fundamentals; developed customized lesson plans resulting in measurable grade improvements.

PROJECTS

FaultClaw – Adversarial Hardware Verification Agent | Python, Verilog, NVIDIA Nemotron

- Built an autonomous multi-agent verification pipeline that ingests an RTL module, generates adversarial test cases via an LLM reasoning agent, executes them against a golden reference model, and produces structured failure reports and coverage metrics without human intervention.
- Architected three coordinated agents: a Spec Reader parsing RTL interfaces into structured representations, an Adversarial Test Generator reasoning across three tiers of test intensity (functional, edge case, and fully adversarial), and a Verification Judge that logs failures and triggers targeted follow-up generation in a closed feedback loop.
- Implemented Breakdown Mode — a full adversarial testing mode that maximizes overflow conditions, stresses boundary values, and chains input sequences to expose latent bugs; demonstrated on a deliberately buggy 4-bit adder caught automatically with precise failure diagnosis.

FSAE Charger Enclosure | Onshape

- Designed a charger enclosure for a Formula SAE electric vehicle, modeling mounting geometry, HV cable routing, ventilation, and isolation features; produced dimensioned 2D manufacturing drawings with tolerances and assembly notes.

RISC-V CPU | Verilog, SystemVerilog

- Implemented a 5-stage pipelined RV32I CPU with hazard detection, data forwarding, ALU, register file, control unit, and memory interface; verified via simulation testbenches and validated on FPGA executing real RISC-V programs.